

Software Requirements Specification (SRS): VitalPath Health Passbook

1. Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for the "VitalPath" mobile application. VitalPath is a centralized "Health Passbook" designed to eliminate the fragmentation of health data by consolidating medication adherence, nutritional routines, physical activity, and professional medical consultations into a single, unified mobile ecosystem.

1.2 Product Scope

VitalPath will serve as a daily-use utility for patients and a management portal for healthcare providers. The scope includes automated activity tracking via hardware sensors, a sophisticated reminder engine for medications and meals, a doctor-patient synchronization bridge, and high-visibility widgets for home-screen adherence tracking.

2. Overall Description

2.1 Product Perspective

VitalPath is a mobile-first solution designed to operate as a middleware between the user's physical health actions and their digital records. It integrates with system-level health APIs (Apple HealthKit/Google Fit) to ensure that the user does not have to manually input data that the phone can already track (like steps or distance).

2.2 User Classes and Characteristics

- **The Primary Patient:** Users ranging from age 18 to 75. They require varying levels of UI simplicity. Their primary goal is adherence to a health regimen.
- **The Healthcare Provider (Doctor):** Professional users who require a high-level overview of patient adherence and the ability to update medical protocols (prescriptions/appointments) remotely.
- **The Caregiver (Secondary User):** Authorized individuals (family/nurses) who monitor the Primary Patient's adherence without full administrative control.

3. User Journeys (Detailed Flow)

3.1 The Patient Journey: From Onboarding to Habit

I. Contextual Onboarding (The Happy Path):

The user opens the app and is greeted by a high-performance splash screen. They authenticate via phone number and Biometrics (FaceID). The app asks for three permissions: Motion (Steps), Notifications (Alarms), and Location (Walking paths). The user then completes a "Health Profile" (Height, Weight, Condition). The journey concludes with the user adding one "Active Medication" to see the dashboard populate.

II. Daily Maintenance:

Every morning, the user receives a "Daily Outlook" notification. They check their "Health Passbook" (The Dashboard) to see their 10,000-step goal and their 8:00 AM pill. As they walk, the progress bar fills in real-time (Antigravity responsiveness). When they take their medicine, they tap a "Log" button directly from the Home Screen Widget or the Lock Screen, never needing to dive deep into menus.

3.2 The Doctor Journey: Remote Management

I. Patient Onboarding:

The doctor logs into the "Provider Portal." They generate a unique 6-digit sync code or QR code and provide it to the patient. Once the patient scans it, the doctor sees the patient's profile appear in their "Active Roster."

II. Protocol Management:

The doctor selects a patient and clicks "Add Prescription." They input the dosage, time, and duration. Upon clicking "Push to Patient," the doctor sees a confirmation that the patient's app has updated. The doctor can also view an "Adherence Report," seeing exactly which doses were missed or rescheduled.

4. System Features & Functional Requirements

4.1 Medicine Management System

- **Inputs:** Name, Unit (mg/ml/pills), Dosage, Frequency (Daily, Weekly, As Needed), Start/End Date, and Image Upload for visual identification.
- **Actions:** Add, Edit, Delete, Skip, and Reschedule.
- **Behavior:** When a medicine is logged, the inventory count decrements. If the inventory reaches a threshold (e.g., 5 pills left), a "Refill Reminder" is triggered.
- **Validations:** Users cannot set an end date that occurs before the start date. Dosage must be a positive numeric value.

4.2 Food Routine & Nutrition

- **Inputs:** Meal Name, Time Window (e.g., 8:00 AM - 9:00 AM), Description, and Nutritional Tags (Low Sodium, Diabetic Friendly).
- **Actions:** Log Meal, Snooze Reminder, Change Routine.

- **Behavior:** The app sends a "Pre-meal" reminder 15 minutes before the scheduled time to help the user prepare.

4.3 Activity Tracking (Steps & Walking)

- **Inputs:** Step Goal (Target), Unit Preference (KM/Miles).
- **Actions:** Start/Stop GPS Walk, Set Daily Goal, View Historical Data.
- **Behavior:** The app uses background processing to pull step counts every 15 minutes. During a "GPS Walk," the app provides real-time haptic feedback every 1 KM reached.

4.4 Doctor-Patient Synchronization

- **Inputs:** Doctor Name, Clinic Name, Contact Number, Appointment Date/Time.
- **Actions:** "Sync with Doctor," "Call Doctor," "View Shared Schedule."
- **Behavior:** The doctor's entries are marked as "Verified." If a user tries to delete a "Verified" entry, a double-confirmation modal appears.

5. Error Handling, Validations, and Edge Cases

5.1 System Validations

- **Network Failure:** If a user logs a medicine while offline, the app stores the timestamp locally in a "Sync Queue." Upon reconnection, it pushes the data to the cloud with the *original* timestamp, not the current one.
- **Input Integrity:** If a user enters a weight that is statistically improbable (e.g., 500kg), the app triggers a soft-validation toast: *"Please double-check your weight entry for accuracy."*

5.2 Edge Cases (The "Designer's Radar")

- **The Timezone Leap:** If a user moves from GMT to PST, the app detects the system clock change. It asks: *"Do you want to keep your 8:00 AM alarm in your home timezone or adapt to local time?"*
- **The Duplicate Log:** If a user logs a dose and then tries to log the same dose within 15 minutes, a critical warning appears: *"Warning: This dose was already logged at [Time]. Taking another may result in an overdose."*
- **Notification Fatigue:** If a user ignores 5 notifications in a row, the app triggers a "Silent Mode Detection" and suggests a different reminder style (e.g., turning on "Critical Alerts").

5.3 Corner Cases

- **Device Storage Full:** If the device cannot save a GPS walk path due to storage, the app must prioritize the "Step Count" (minimal data) and discard the "GPS Path" (large data) to ensure the core metric is saved.
- **Simultaneous Updates:** If the Doctor updates a schedule at the exact moment the User is editing it, the "Server-Side Stamp" wins, but the user is notified: *"Your doctor just updated this routine. Your recent changes were overwritten to match the medical plan."*

6. Non-Functional Requirements

6.1 Performance (Antigravity Standards)

- **Latency:** Every button press must result in a UI change within **100ms**.
- **Fluidity:** All transitions between the Dashboard and Detail screens must run at **60fps** (or 120fps on ProMotion displays).

6.2 Security & Privacy

- **Encryption:** All health data must be encrypted with **AES-256**.
- **Compliance:** The system architecture must follow **HIPAA** and **GDPR** guidelines, ensuring users have the "Right to be Forgotten" (Full data wipe).

7. Design Visual & Interactions

- **The Dashboard:** A vertical timeline. Completed tasks move to a "Faded" state; upcoming tasks are "Elevated."
- **Widgets:** Support for "Lock Screen Widgets" (iOS 16+) showing the next medicine time at a glance.
- **Haptics:** Distinct vibration patterns for "Medicine" (Short, sharp) vs. "Steps Goal Reached" (Long, celebratory).

This document provides the blueprint. From here, we would move into **High-Fidelity Wireframing** and **Prototyping** the specific transitions between the Dashboard and the Doctor's Portal.

How would you like to handle the "Refill" logic? Should we integrate with local pharmacy APIs, or keep it as a simple manual inventory tracker for version 1.0?